

Beyond AUC: A Cost-Sensitive, Calibration-First Evaluation of Card-Not-Present Fraud Models under Temporal Drift

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Abstract

Card-not-present (CNP) fraud detection is usually reported as a ranking problem: a model, a shuffled cross-validation, an AUC. Production fraud systems answer a different question — *which transactions should be declined, at what dollar cost, next month* — and the usual evaluation answers it badly. Using the IEEE-CIS dataset (590,540 e-commerce transactions, 3.50% fraud), we build a deliberately standard model stack (regularized logistic regression, LightGBM) and spend the evaluation budget where production risk actually lives. Under a strict out-of-time protocol (train months 0–3, validate on month 4, touch month 5 once), random 5-fold cross-validation overstates test-month performance by **0.30 PR-AUC and 22 points of TPR@5%FPR** for the gradient-boosted challenger (0.87→0.56; 0.89→0.67) while barely moving the linear champion — the optimism tracks a model's capacity to memorize entities. A two-parameter cost model (missed fraud costs the amount; a false decline costs $k \times$ amount) admits an amount-independent optimal threshold $p^* = k/(1+k)$ — but only on calibrated probabilities. On the test month the analytic threshold on calibrated scores beats the F1-optimal threshold by **+7.9 points of savings** (paired 95% CI +5.1 to +10.8) and cost-thresholding the raw class-weighted scores by +2.2 points (\$11.1K on the month; paired CI −0.2 to +4.3). The final system attains **39.4% cost savings** against the best trivial policy at the central $k = 0.15$ (95% CI 35.8–42.8%), with TPR at the 5%-FPR operating point of **67.2%** (65.5–68.8%). Weekly decay curves, PSI, and adversarial validation quantify drift: monthly retraining recovers ~7.5 points of the savings the static model loses in month 5.

1. Introduction

The dominant public benchmark treats fraud detection as pure ranking, but the economics of CNP payments are decision-theoretic and sharply asymmetric. Javelin Strategy & Research estimated falsely declined card transactions at almost **\$118 billion per year** in the U.S. — with 15% of cardholders experiencing a false decline within a year and 39% of those abandoning the declined card afterward [4]. The problem has not aged away: in PYMNTS Intelligence's 2026 merchant survey, **85% of merchants** call reducing friction for legitimate customers their biggest fraud-prevention challenge, and nearly half estimate that **up to 5% of legitimate orders** are incorrectly declined, roughly \$50 billion of industry revenue [5]; the 2024 MRC/Visa

Acceptance global fraud report finds most merchants self-report false-positive rates of 2–10% of orders [3]. Meanwhile the threat side keeps industrializing — Visa's Spring 2025 Biannual Threats Report tracks a 22% six-month rise in enumerated transactions and attributes ~US\$1.1B of annual follow-on fraud to enumeration alone [7].

This tension — fraud loss vs. false-decline loss — is precisely the objective Visa assigns to its own CNP scoring product: Visa Deep Authorization is described as a deep-learning risk score built to "boost approvals for card-not-present transactions" while detecting and declining fraud [6]. A model that improves AUC but is thresholded naively, on miscalibrated scores, can *lose* money relative to a weaker model with an honest decision layer. That decision layer, not the classifier, is this project's subject.

We make four contributions on the public IEEE-CIS dataset [1]:

1. **A cost-sensitive decision layer evaluated in dollars.** With FN cost = amount and FP cost = $k \times \text{amount}$, expected-cost decisioning declines iff $p > k/(1+k)$ (Elkan [2]) — an amount-independent threshold we derive in one line, stress across $k \in \{0.05, \dots, 1.0\}$, and compare against F1-optimal, Youden-J, and trivial policies using Bahnsen-convention savings [10], with bootstrap confidence intervals, on a strictly out-of-time test month.
2. **A measured optimism gap.** The same models, features, and hyperparameters are evaluated under random 5-fold CV, GroupKFold by month, and the temporal protocol; the difference between the first and last is the optimism a leaderboard-style evaluation silently adds.
3. **Calibration as a load-bearing component, priced in dollars.** Class-weighted training deliberately distorts probabilities; we repair them with Platt/isotonic calibrators fit only on the validation month [12], and price the uncalibrated-vs-calibrated difference at the analytic threshold.
4. **Drift made operational.** Weekly out-of-time decay under three retraining cadences (static / expanding / sliding), PSI on the model's top features, and adversarial validation — the monitoring quantities a model-risk owner would actually track.

Everything is CPU-reproducible (`make all`), every aggregation feature is provably causal (property-tested), and the test month is touched exactly once.

2. Related work

Cost-sensitive learning. Elkan's foundational treatment shows that for a binary decision with per-example costs, the optimal decision on calibrated probabilities is a threshold determined by the cost matrix alone [2]. Bahnsen, Aouada & Ottersten develop example-dependent cost-sensitive methods for card fraud and introduce the *savings* convention used here — cost

improvement over the best trivial (accept-all / decline-all) policy [10]. We deliberately keep the model conventional and put the cost-sensitivity in the decision layer, where it can be audited.

Calibration. Platt scaling fits a sigmoid on a held-out score sample [12a]; isotonic regression fits a monotone step function and needs more data but no shape assumption (Zadrozny & Elkan [12b]). Guo et al. document how modern learners are systematically miscalibrated and how simple post-hoc maps repair them [12c]. Fraud adds two twists: class-weighting (used here) shifts scores by design, and drift decays calibration over time — we track weekly Brier/ECE out-of-time for exactly that reason.

IEEE-CIS top solutions. The winning entry (Deotte & Yakovlev) reconstructed a client identifier from `card1`, `addr1` and `D1` and aggregated behavior over it, validating with GroupKFold using months as non-overlapping groups because client-level memorization otherwise masquerades as generalization [8]. We cite it as *precedent and warning*: the same mechanics that win a leaderboard are the ones that leak under a shuffled split. Our aggregates are restricted to expanding, past-only statistics, and no label-derived aggregate is used at all (production chargeback labels arrive weeks late).

Drift and operating points. The Bank Account Fraud benchmark (Jesus et al., NeurIPS 2022) builds dynamic, imbalanced tabular datasets and standardizes evaluation at a fixed operating point — "we select the threshold in order to obtain 5% false positive rate (FPR), and measure the true positive rate (TPR) at that point" [11]. We adopt TPR@5%FPR as the headline ranking metric and add the dollar-denominated layer on top.

3. Data

The IEEE-CIS Fraud Detection corpus (Vesta Corporation, 2019) contains **590,540** e-commerce transactions over **182 days**, of which **20,663 (3.499%)** are fraudulent; **24.4%** of transactions have device/identity enrichment in a second table joined on `TransactionID` [1]. Feature blocks: core transaction fields (amount, product code, card, address, distance, purchaser/recipient email domains), counts `C1-C14`, time deltas `D1-D15`, match flags `M1-M9`, 339 anonymized Vesta-engineered features `V1-V339`, and 41 identity columns. `TransactionDT` is a second-offset from an undisclosed reference; we define `day = TransactionDT // 86400`, `week = day // 7`, `month = min(day // 30, 5)` (the 2–3 day tail stub is merged into the final month; DECISIONS D-003).

Two facts shape everything downstream. First, the stream is **non-stationary**: monthly fraud rate moves from 2.48% (month 0) through 4.04% (month 1) down to 3.47% (month 5), and transaction volume swings weekly (Figure 1, Figure 2). Second, the **label is operational, not oracular**: `isFraud` reflects Vesta's chargeback-linked labeling; downstream transactions of a flagged account/card can inherit the label, and reporting delay censors the newest data. Both properties argue for chronological evaluation and against label-based feature aggregation. All

modeling below uses the labeled training file only; Kaggle's unlabeled test set appears nowhere in any evaluation claim.

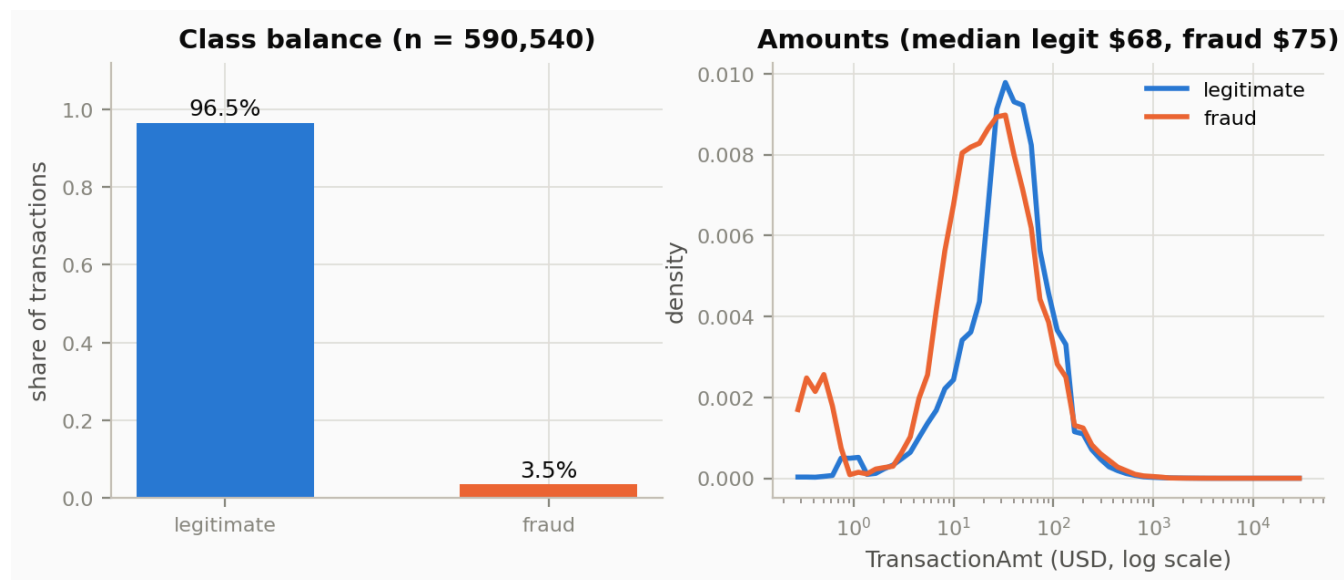


Figure 1 — class balance and amount distributions

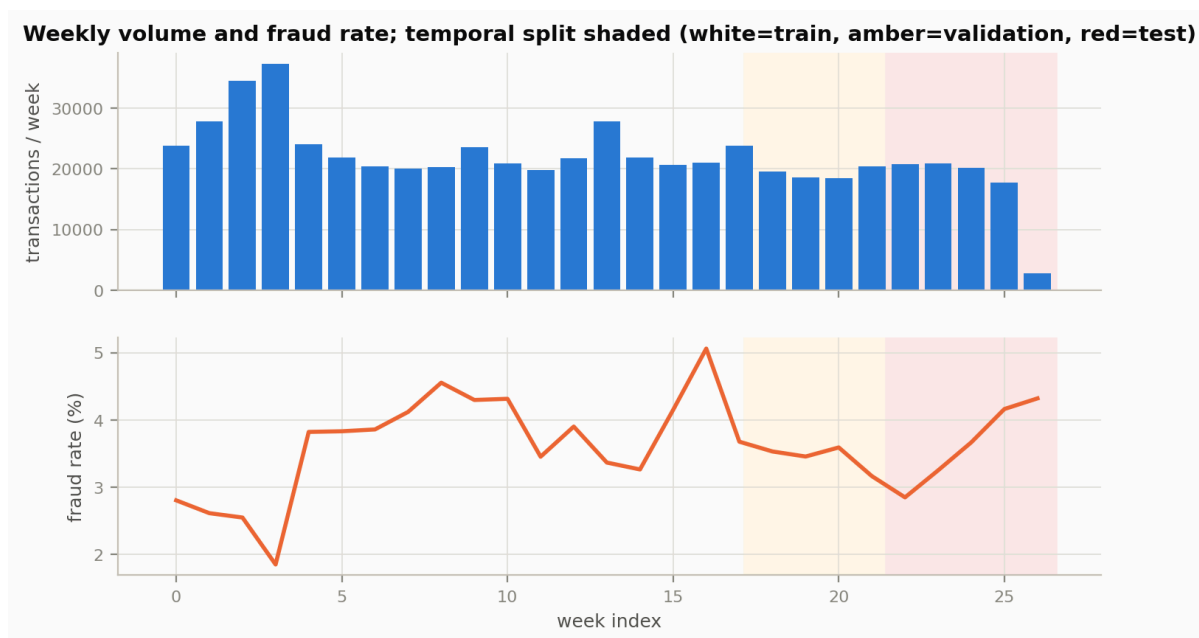


Figure 2 — weekly volume and fraud rate, with the temporal split

4. Methodology

4.1 Temporal protocol

The fixed temporal protocol: every choice happens upstream of the test month

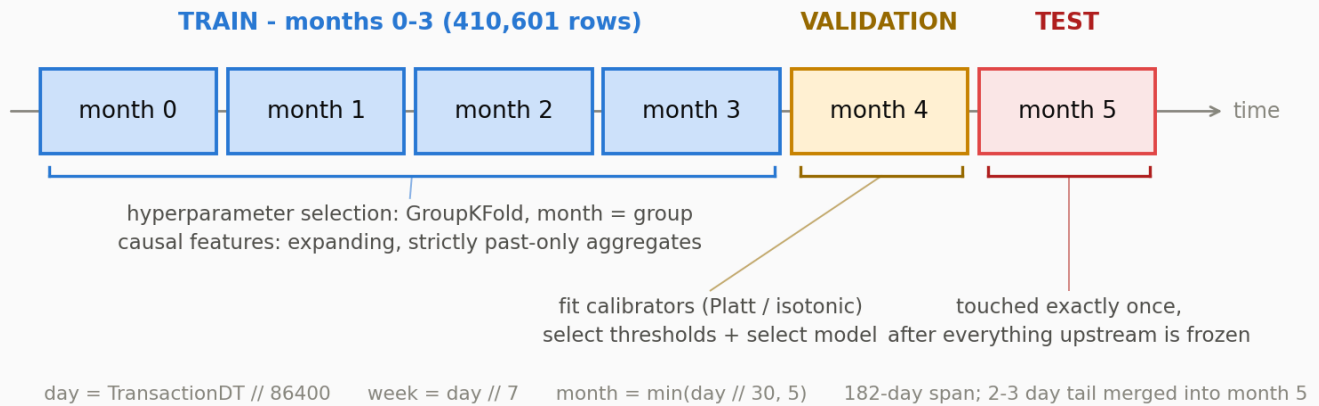


Figure P — the fixed temporal protocol

Train: months 0–3 (410,601 rows). Validation: month 4 (85,303 rows) — used for calibration fitting, threshold selection, and model selection. Test: month 5 (94,636 rows) — evaluated exactly once, after every upstream choice is frozen. Hyperparameters are selected with GroupKFold using month as the group within the training window. Weekly curves use `week = day // 7`. Automated tests assert split integrity (no temporal overlap; fold groups disjoint) and feature causality (see 4.3).

4.2 Models and imbalance

The **champion** is L2-regularized logistic regression on interpretable blocks (core, counts, time deltas, match flags, numeric identity), with window-fit medians/scaling, missingness indicators for the D block, and top-30 one-hot vocabularies fit on the training window only. The **challenger** is LightGBM on all blocks with native categorical handling. Imbalance is handled with `scale_pos_weight` (= neg/pos of each fit window); we do not use SMOTE or any synthetic oversampling, because resampling distorts the probability scale that the decision layer depends on and interacts badly with post-hoc calibration — weighting plus explicit recalibration achieves the same operating-point control while keeping the probability semantics auditable (DECISIONS D-004/D-005). Search is deliberately modest (Optuna TPE, 12 trials inside a 45-minute budget, GroupKFold-by-month folds; best fold PR-AUC 0.626); rigor is spent on evaluation, not tuning depth.

4.3 Causal features

Beyond row-local transforms (log-amount, cents pattern, hour, day-of-week), we add expanding **past-only** aggregates per `card1` and per `(card1, addr1)`: prior-transaction count, seconds since the entity's previous transaction, past mean amount, and current/past-mean amount ratio. `tests/test_causal_features.py` enforces the defining property by *prefix recomputation*: the feature value at row *i* is identical when every later row is deleted. No aggregate uses `isFraud` —

chargeback labels arrive with multi-week delay in production, and past-label features would silently assume instant label availability.

4.4 Cost framework

Let a transaction have amount A and fraud probability p . Approving a fraud costs A (write-off proxy); declining a legitimate order costs kA (lost margin plus attrition proxy). Expected costs: approve $\rightarrow pA$; decline $\rightarrow (1-p)kA$. Decline exactly when

$$pA > (1-p)kA \iff p > k/(1+k),$$

and A cancels: the optimal threshold is *amount-independent* — but the inequality is a statement about a **probability**, so it is only actionable on calibrated scores. This is why calibration is load-bearing rather than cosmetic. Central case $k = 0.15$ (threshold 0.1304); sensitivity grid $k \in \{0.05, 0.15, 0.30, 0.60, 1.0\}$. Savings follow Bahnsen et al. [10]: `savings = 1 - Cost_model / Cost_baseline` with `Cost_baseline = min(cost(approve-all), cost(decline-all))`. Six policies are compared on identical test data: approve-all, decline-all, F1-optimal and Youden-J thresholds (selected on validation), and the analytic cost threshold applied to uncalibrated vs. calibrated probabilities. Platt and isotonic calibrators are fit on month 4 only; the method with lower validation Brier is frozen for test.

4.5 Drift and ablation design

Weekly decay is evaluated over months 4–5 for three cadences — **static** (train once on months 0–3), **expanding** (retrain monthly on all past), **sliding** (trailing 3 months) — under a uniform rule: every window holds out its final 14 days for calibration (D-006). Input drift is tracked with PSI (train window $\rightarrow$ each evaluation week) for the model's top-20 gain features, plus adversarial validation (a classifier separating train rows from out-of-time rows; $AUC \approx 0.5$ means indistinguishable), run both with and without the cumulative aggregate features. Incremental validity (Stage 6) retrains the challenger under fixed hyperparameters and tree count on forward-addition ($A \rightarrow A + \dots + F$) and leave-one-block-out configurations, scoring $\Delta TPR@5\%FPR$ and Δ savings on validation; the selected configuration is confirmed once on test.

5. Results

This section reports the persisted outputs in `results/metrics/`; every figure is generated by `make figures` from those artifacts.

5.1 The optimism gap

Identical features and tuned hyperparameters, three split regimes:

LightGBM	ROC-AUC	PR-AUC	TPR@5%FPR
random 5-fold CV (mean)	0.968	0.866	0.891
GroupKFold by month (mean)	0.917	0.618	0.700
out-of-time test (month 5)	0.904	0.564	0.672
gap: random – test	+0.064	+0.302	+0.219

The logistic champion's corresponding gaps are +0.020 / +0.017 / +0.035. Two observations matter. First, the shuffled estimate is not mildly optimistic — it reports a model roughly *30 PR-AUC points* better than the one that exists next month, and the distortion is largest exactly where fraud metrics live (the high-precision region; ROC-AUC hides it). Second, the gap is model-dependent: the boosted model, which can memorize card-profile behavior across a shuffled split, inflates; the linear model, which cannot, does not. Grouped-by-month CV lands within 3–5 points of the true out-of-time numbers and is the honest offline surrogate. **Takeaway (Figure 3): the choice of split changes the reported number more than any modeling choice in this paper.**

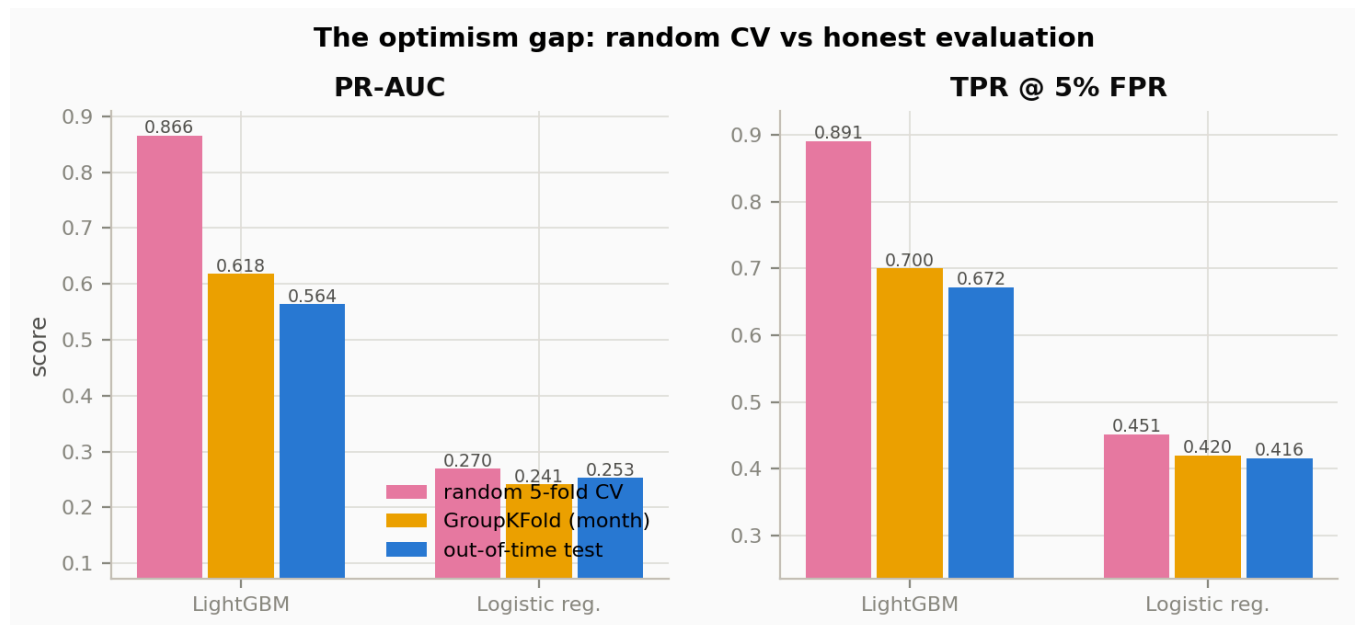


Figure 3 — random-split vs honest evaluation

5.2 Calibration

Class-weighted training (`scale_pos_weight` ≈ 27) leaves the raw scores usable for ranking but distorted as probabilities (Platt slope 0.67 on validation: raw log-odds are overconfident). Isotonic regression, fit on month 4 only, wins on validation Brier (0.01802 vs 0.01820 Platt vs 0.01866 raw) and transfers to the untouched test month: Brier 0.0217 $\rightarrow$ 0.0212, log-loss 0.1053 $\rightarrow$ 0.0884, ECE 0.0146 $\rightarrow$ 0.0041. Weekly out-of-time tracking shows the calibrated scores hold their reliability across both evaluation months. **Takeaway (Figure 4): a one-month held-out**

calibrator repairs the probability scale that the decision layer is about to lean on — cheaply and stably.

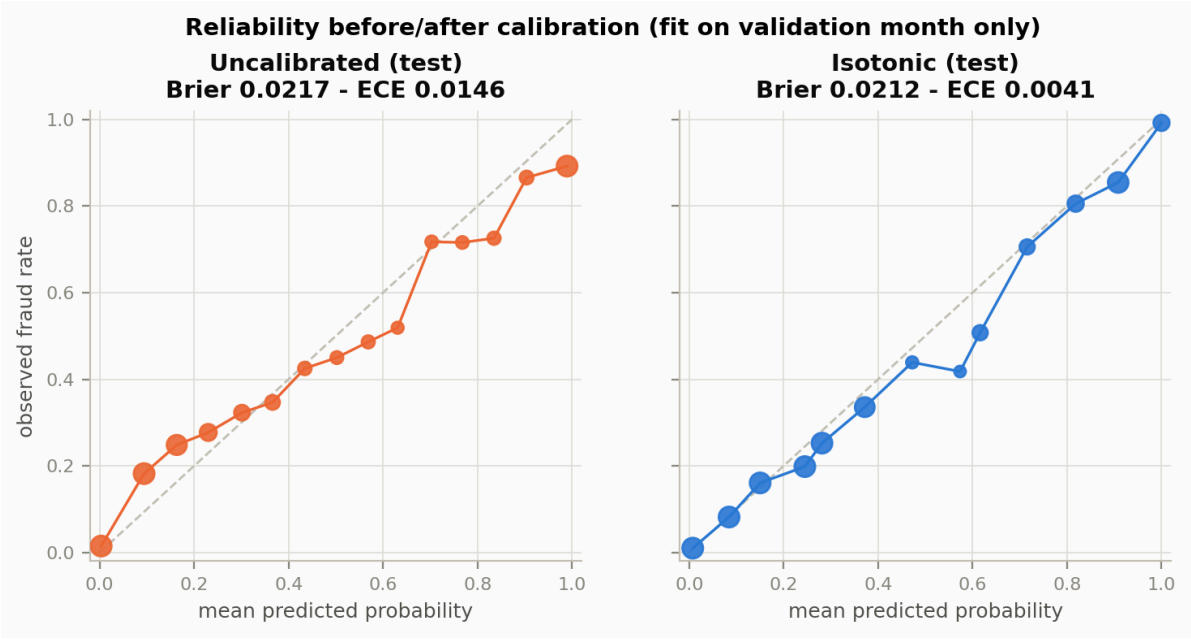


Figure 4 — reliability before/after calibration

5.3 The decision layer in dollars

Test month: 94,636 transactions, 3.47% fraud; baseline policy is approve-all (cost \$495,244). All thresholds frozen on validation.

policy	total cost	savings	fraud \$ caught	legit \$ declined	TPR	FPR
approve all	\$495,244	0.0%	\$0	\$0	0.000	0.000
decline all	\$1,872,497	−278.1%	\$495,244	\$12,483,317	1.000	1.000
F1-optimal	\$339,675	31.4%	\$171,157	\$103,921	0.439	0.007
Youden J	\$359,583	27.4%	\$367,605	\$1,546,285	0.748	0.089
cost thr., uncalibrated	\$311,403	37.1%	\$221,430	\$250,591	0.521	0.015
cost thr., calibrated	\$300,271	39.4% [35.8, 42.8]	\$274,003	\$526,866	0.604	0.029

The ordering is the argument. Youden J catches the most fraud dollars but burns \$1.55M of legitimate volume to do it; F1 is precise but timid; the analytic threshold $k/(1+k)$ beats both — **+7.9 points of savings over F1** (paired 95% CI [+5.1, +10.8], positive in 100% of resamples). Calibrating before thresholding adds +2.2 points / \$11.1K on the month (2.25% of baseline); the paired CI [−0.2, +4.3] just crosses zero, so on a single test month this increment is directionally

consistent (96% of resamples) rather than individually significant — we report it as such. Across the k grid (Figure 6), the analytic policy adapts and stays positive everywhere (58.6% savings at $k = 0.05$, 19.3% at $k = 1.0$), while the fixed Youden threshold collapses to **-238%** at $k = 1.0$: a threshold chosen without reference to costs does not merely underperform, it changes sign. **Takeaway (Figure 5): where you cut matters more than how well you rank — and the right cut is a formula, not a grid search, once probabilities are calibrated.**

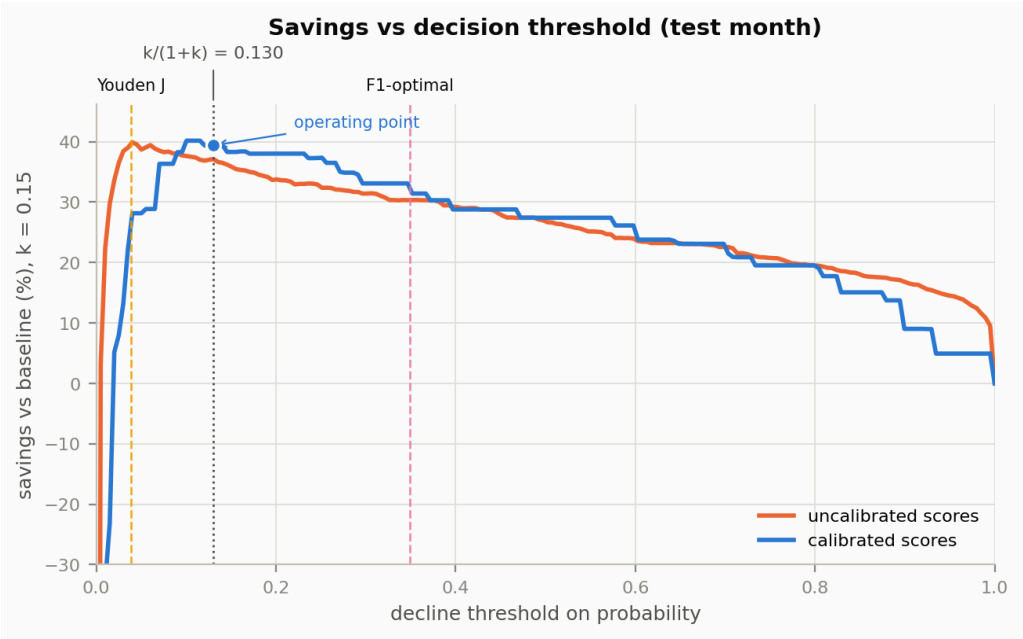


Figure 5 — savings vs threshold

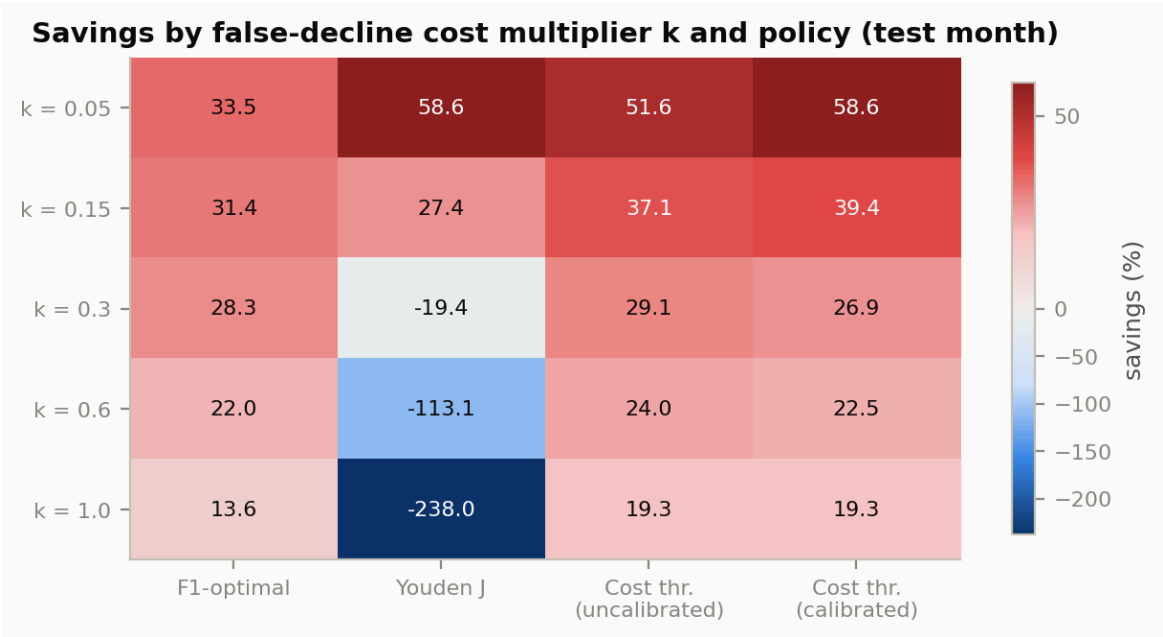


Figure 6 — sensitivity across k

5.4 Drift and retraining cadence

Weekly performance decays visibly within the two out-of-time months, with a trough at week 20 and only partial recovery. Retraining at the month boundary changes the picture (month-5 means, uniform holdout-calibration protocol):

cadence	TPR@5%FPR	PR-AUC	savings (k=0.15)
static (train once)	0.651	0.535	0.354
expanding (all past)	0.703	0.600	0.429
sliding (last 3 months)	0.696	0.597	0.422

One monthly retrain recovers **+7.5 points of savings** over the static model; keeping all history edges out the 3-month window, so recency matters less than freshness. On the input side, PSI flags concentrated, interpretable drift: `id_31` (browser version) reaches PSI 1.36 — software-ecosystem churn, invisible to a static model — and the cumulative entity counters drift mechanically (0.37); 3 of the top-20 features breach the conventional 0.2 alarm by the final week. Adversarial validation separates train from out-of-time rows with AUC 0.964 — and still 0.899 after removing the mechanically-trending aggregate features, so the shift is genuine covariate drift, not an artifact of the counters. **Takeaway (Figures 7-8): this dataset retires "train once, deploy forever" on its own — and the decision-relevant decay (savings) is steeper than the rank-metric decay.**

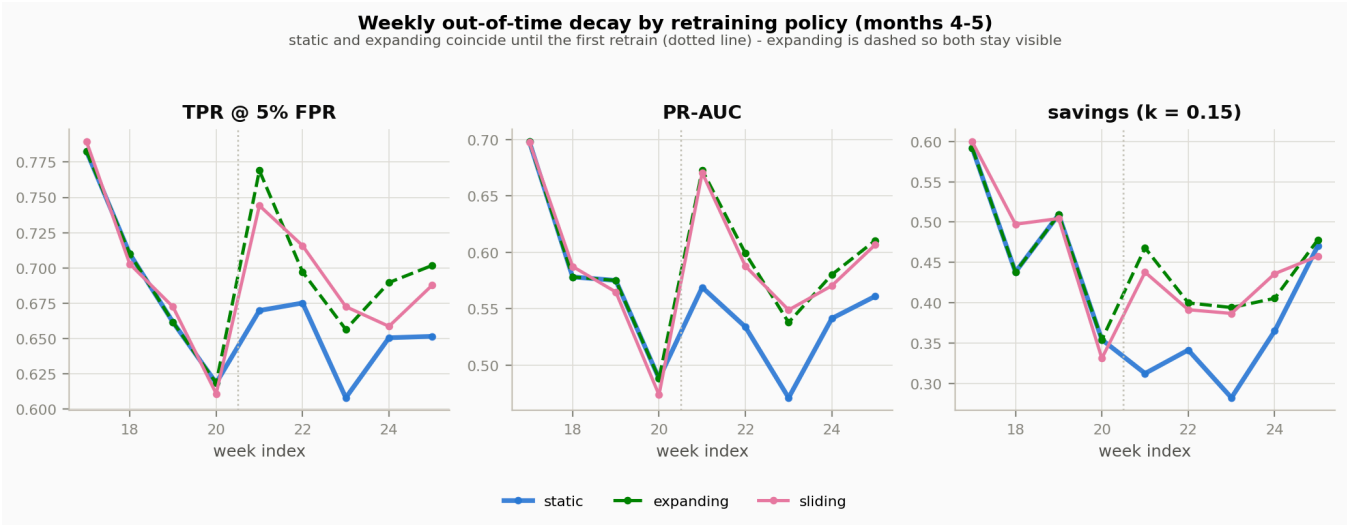


Figure 7 — weekly decay by retraining policy

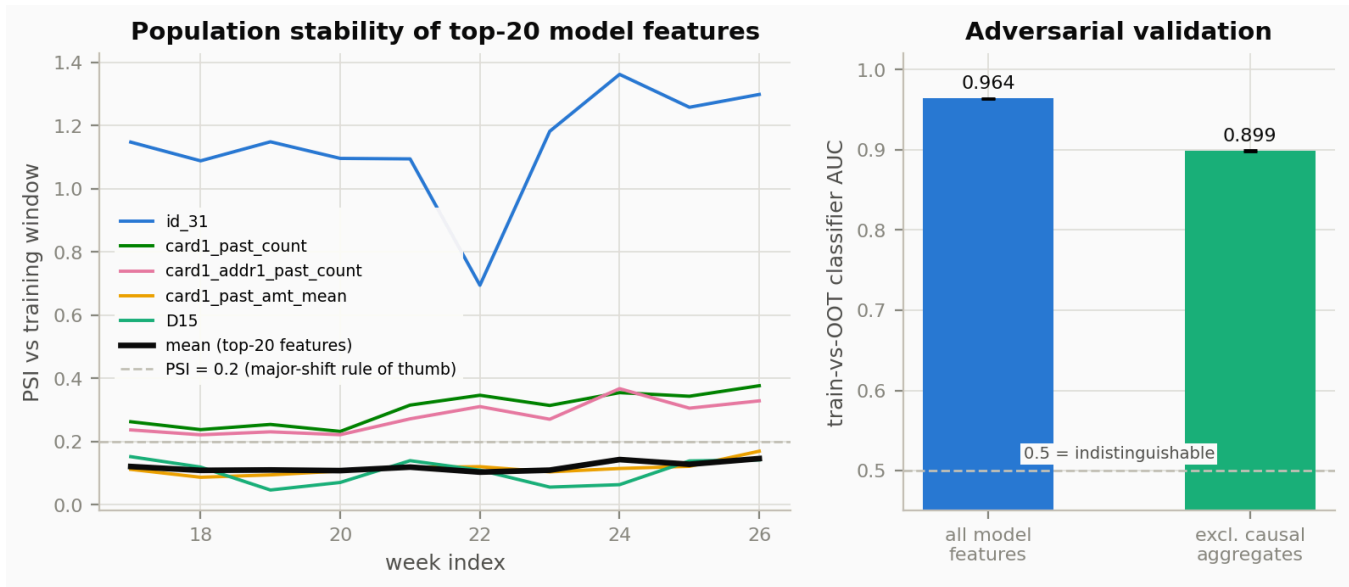


Figure 8 — PSI and adversarial validation

5.5 Incremental validity of the feature blocks

Forward addition on validation (fixed hyperparameters and tree count):

configuration	TPR@5%FPR	savings
A core (+ causal aggregates)	0.439	0.262
+ B counts (C1-C14)	0.616	0.405
+ C time deltas (D1-D15)	0.656	0.432
+ D match flags (M1-M9)	0.666	0.449
+ E Vesta (V1-V339)	0.676	0.473
+ F identity	0.685	0.470

Core + counts — 40 interpretable features — deliver **86% of the full configuration's savings**. The 339 anonymized Vesta features buy +2.4 points of savings; the identity table buys +0.9 points of TPR while costing −0.3 points of savings at this operating point. Leave-one-block-out confirms no block strictly harms both criteria (removing counts costs −4.9 points of savings; removing match flags is cost-neutral), so the full configuration is retained for the headline model. **Takeaway (Figure 9): most of the black-box's dollar value lives in a small auditable core — a governance-relevant fact when every extra block is attack surface, latency, and model-risk documentation.**

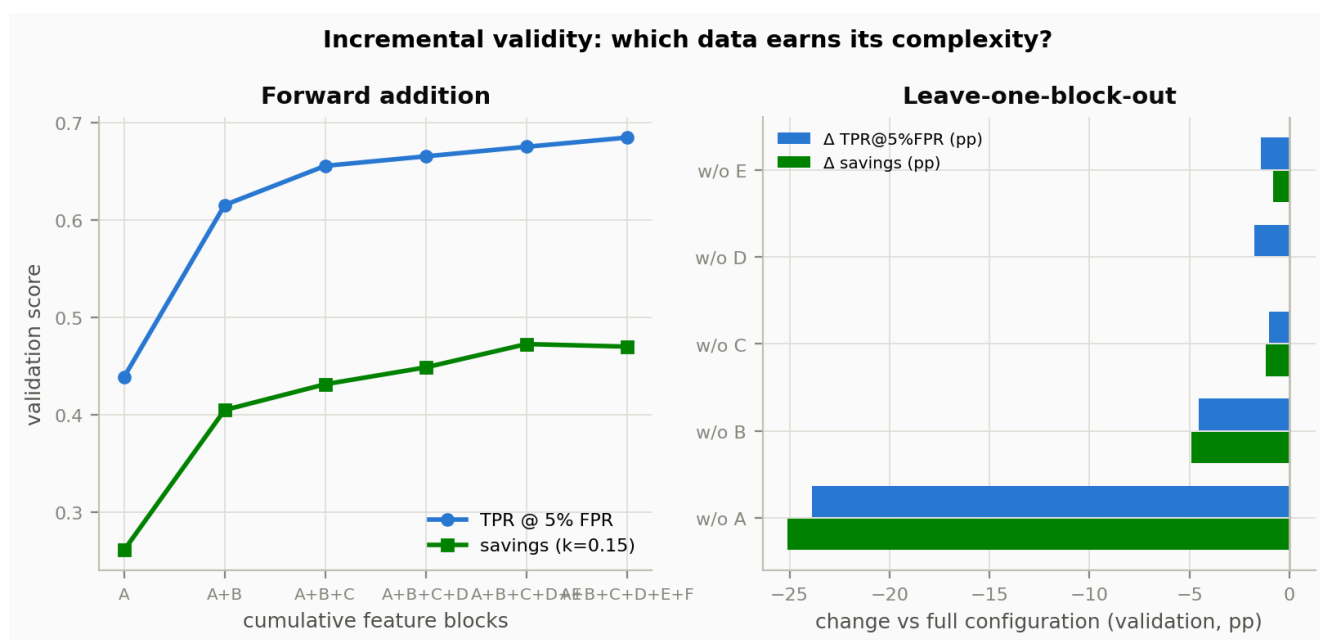


Figure 9 — forward addition and leave-one-block-out

5.6 Explainability

TreeSHAP on a stratified 20K-row test sample puts velocity and identity-consistency signals at the top: C13 and C1 (Vesta count features), purchaser email domain, and — notably — three of the causal aggregates built here (`card1_past_amt_mean`, `card1_addr1_past_count`, `card1_past_count`) rank inside the top eleven. The paired waterfalls show the decision anatomy a dispute analyst needs: the true positive is declined on a first-seen card profile transacting at an unusual amount; the false positive is a legitimate customer who *looks* first-seen — exactly the failure mode that step-up authentication, rather than a hard decline, should absorb. **Takeaway (Figure 10): the model's dollars come from behavioral-consistency features, which is also where its false declines come from.**

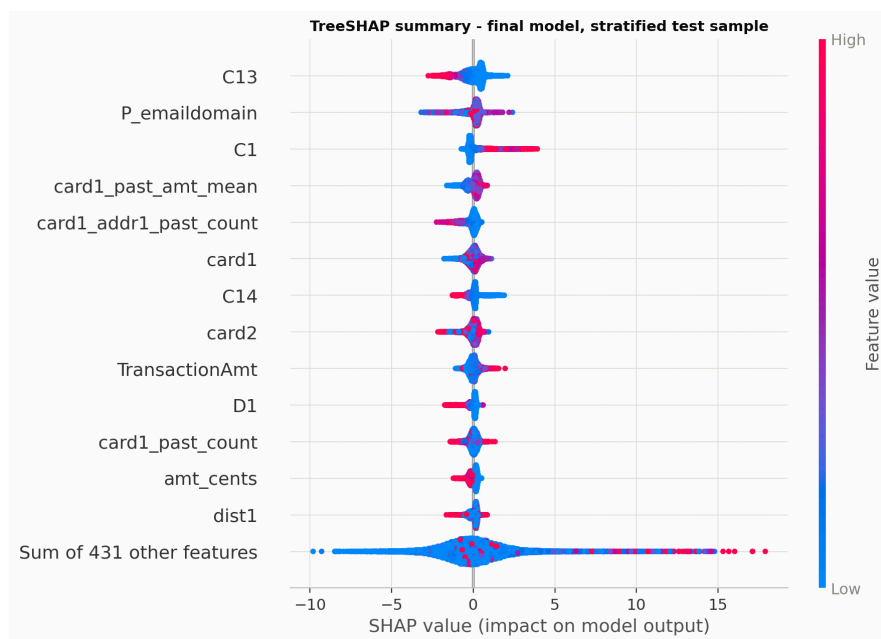


Figure 10a — TreeSHAP beeswarm

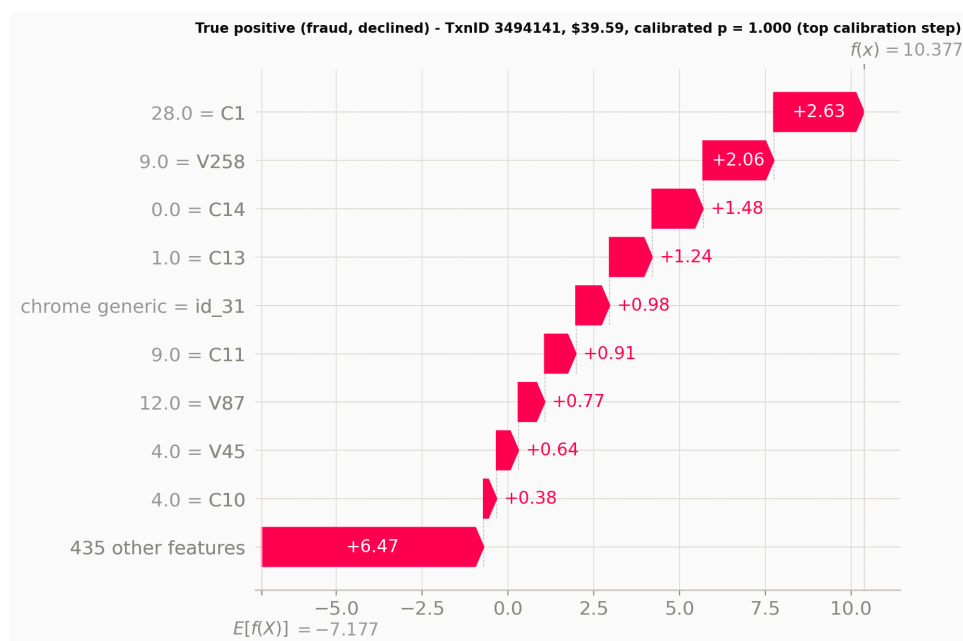


Figure 10b — true positive waterfall

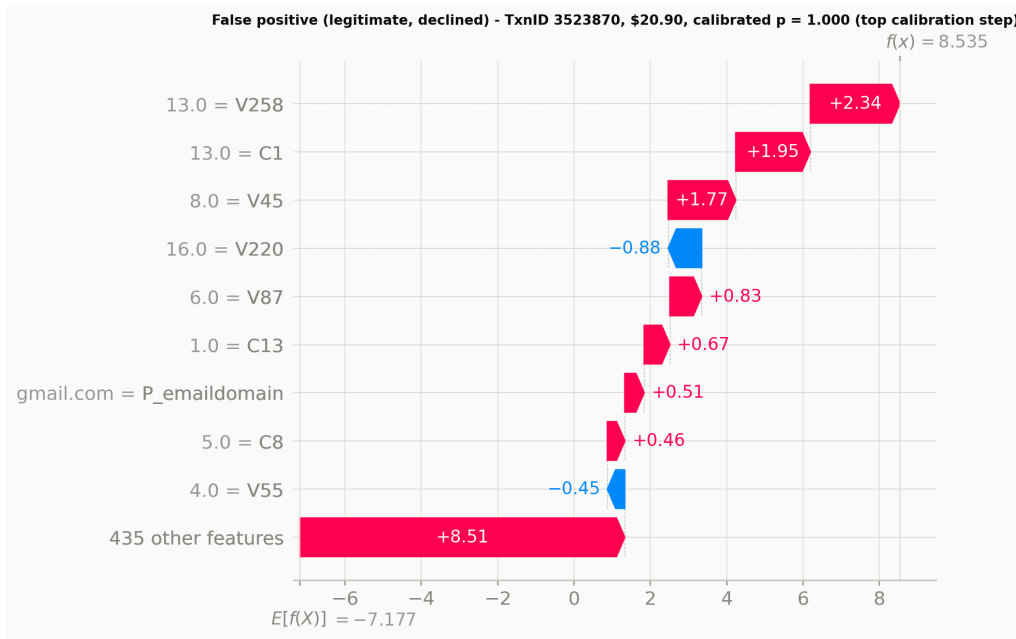


Figure 10c — false positive waterfall

6. Discussion

Limitations. This is one anonymized 2019 e-commerce dataset; the drift studied is historical, not adversarially live. The label is chargeback-linked and inherits reporting noise, delay, and within-account propagation — our "ground truth" is itself an operational artifact. The cost model is a two-parameter proxy: real FN costs include interchange, disputes, and recovery; real FP costs are heterogeneous across customers and partially recoverable through step-up authentication rather than hard declines; k is best read as a governance dial, which is why we report the full sensitivity grid rather than defend a single value. Calibrators and thresholds share the validation month; with more months one would separate those roles.

What changes at Visa scale. An issuer-side CNP score runs at authorization time under a roughly millisecond latency budget, which constrains feature stores to streaming, incrementally-updatable state (our expanding aggregates are the batch shadow of that design). Decisions feed back into the data: declined transactions never earn labels, creating selective label censoring that offline benchmarks cannot exhibit. Adversaries adapt to the deployed boundary — enumeration attacks alone shifted 22% in six months [7] — so the drift measured here is a lower bound on the operational problem. And at network scale, calibration is not a nicety but the contract: issuers consume scores as probabilities when setting strategy thresholds.

A monitoring plan this analysis directly supports. (1) Weekly TPR@5%FPR and savings on a rolling labeled window, alarmed against the decay slopes in Figure 7. (2) Weekly Brier/ECE on the deployed calibrator; recalibrate when the uncalibrated-vs-calibrated dollar gap reopens. (3) PSI on the top-20 features with the conventional 0.2 trigger, plus a quarterly adversarial-

validation AUC as an omnibus shift detector. (4) Retraining cadence set by the measured curves — retrain when its projected savings recovery exceeds retraining and review cost, not on a calendar reflex. (5) Champion-challenger with the logistic champion as a sanity floor and threshold changes governed as decisions, not tuning.

7. Reproducibility

Python 3.11; pinned versions in `requirements.txt`; seed 42 throughout. Data (Kaggle credentials + accepted competition rules required):

```
kaggle competitions download -c iieee-fraud-detection -p data/raw/  
# or place iieee-fraud-detection.zip at data/raw/ manually  
make data      # checksum, unpack, schema validation report  
make features  # causal feature build + EDA artifacts  
make train     # tuning + three split regimes + final model + scores  
make evaluate  # calibration, cost layer, ablation, drift, single test pass  
make figures   # all paper figures from persisted artifacts  
make test      # split-integrity, causality, and cost-model tests  
make notebooks # generate + execute the six thin notebooks  
make app       # streamlit demo from cached artifacts (no Kaggle needed)
```

Raw data is never committed. SHA-256 checksums of the archive and CSVs are recorded in `results/metrics/data_checksums.json`. Wall-clock on the 2-vCPU build machine: `make data + features` $\approx$ 6 min; `make train` $\approx$ 2 h 03 m (45 m of it the capped tuning budget); `make evaluate` $\approx$ 2 h 33 m (ablation 55 m, drift 31 m, final test pass 67 m — of which TreeSHAP on 20K rows $\times$ 2,190 trees is 65 m); figures + notebooks $\approx$ 3 m. Total $\approx$ **4 h 45 m**; on a typical 4–8-core laptop LightGBM's near-linear thread scaling brings this into the 2–3 h range. The Streamlit demo runs entirely from `app/artifacts/` (a cached, stratified 20K-row test sample); deployment notes for Streamlit Community Cloud / Hugging Face Spaces are in `app/README-deploy.md`, including the competition-rules caveat on redistributing data-derived artifacts.

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Label caveat, dataset licensing, and the full decision log: `DECISIONS.md`. The dataset is subject to the Kaggle competition rules and is not redistributed by this repository.